Analysing COVID-19 Data with R

The COVID-19 pandemic has taken the world by storm. Every nation on earth has joined the fight against this vicious virus, and every little contribution matters. The Internet has also provided the means to effectively use statistical and analytic tools to study the disease and its spread, thereby helping us to understand the implications of the pandemic. Data analysis is an effective tool to study the spread of the virus. This article outlines how to analyse COVID-19 data using R.

Data analysis helps us to understand complex data, identify the trends of data growth, and predict values for some distant points. Exploration of data provides the means to identify the pattern of its distribution and to discover its statistical model. As COVID-19 data is available on many reliable websites, it is possible to explore it for different statistical data analyses and modelling.

The objective of this article is to give readers an idea of data analysis on use cases. Interested readers can initiate exploratory analysis on this data for further understanding and thereby contribute to the welfare of human beings during this pandemic.

Data acquisition

There are many well documented and organised COVID-19 data repositories. For instance, the GitHub repository of Johns Hopkins University Centre for Systems Science and Engineering (JHUCSSE) is highly reliable and easily accessible. From the URL, one can collect the latest data on the worldwide spread of COVID-19. Readers can download or read the data directly from the URL.

```
library(data.table) df <-
fread("https://raw.githubusercontent.
com/CSSEGISandData/COVID-19/master/
csse_covid_19_data/csse_covid_19_time_
series/time_series_covid19_confirmed_
global.csv")
df <- data.frame(df,header=TRUE,strings
AsFactors=FALSE)
```

Preprocessing

Preprocessing of the acquired data is an essential step of data analysis. Proper processing helps us to organise data efficiently and reduces the chances of errors in the final exercise. Preprocessing also makes the data complete by filling in the missing data and removing odd values.

Here we shall check the data for NA and fill in the blank *Province. State* fields with *Country.Region* values. As it is a huge data set, we shall demonstrate the analysis on some selective countries, namely, China, India, Pakistan and Iran.

```
library(dplyr)
df$Province.State <- sub("^$", NA,
df$Province.State)
df <- transform(df, Province.
State = ifelse(Province.State=="" |
is.na(Province.State), Country.Region,
Province.State))
#Replace all fields with 0 if the value
is NA
df <- df %>% replace(.=="NA", 0) #
replace with 0
df <- df[with(df, Country.Region %in%
c("China","India","Pakistan","Iran")),]
```